Ans. (b) In the passage, it has been argued that some crops may respond favorably to climate change while others may have poor productivity because of climate change.
A plan can be made to shift the locations of existing crops based on changes made in water availability and temperature stability at different locations.
This will ensure high productivity and adoptability of a agricultural system to climate change.
Hence, the correct answer is 2 and 3 only.

7. According to the passage, why is it important to promote agricultural research in India ?
(a) To predict variations in monsoon patterns and to manage water resources .
(b) To make long term investment decisions for economic growth .
(c) To facilitate wider adaptability of crops
(d) To predict drought conditions and recharge aquifers.

Ans. (c) According to the passage some crops may respond favorably to the changing conditions other may not. This emphasizes the need to promote agricultural research and create maximum flexibility in the system to permit adaptations and thus India needs to make long term investment in research and development in agriculture.

Passage-2

It is essential that we mitigate the emissions of greenhouse gases and thus avoid some of the worst impacts of climate change that would take place in coming years and decades. Mitigation would require a major shift in the way we produce and consume energy. A shift away from overwhelming dependence on fossil fuels is now long overdue, but unfortunately, technological development has been slow arid inadequate largely because government policies have not promoted investment in research an development, myopically as a result of relatively low prices of oil . It is now, therefore, imperative for a country like India treading the opportunity of harnessing renewable energy on a large scale as a national imperative. This country is extremely well endowed with solar, wind and biomass courses of energy. Where we have lagged, unfortunately, is in our ability to develop and to create technological solutions for harnessing these resources.

One particular trajectory for carry in out stringent mitigation of greenhouse gas emissions assessed by the Inter-governmental Panel on Climate Change (IPCC) elderly shows the need for ensuring that global emissions of greenhouse gases peak no later than 2015 and reduce rapidly thereafter. The cost associated with such a trajectory is truly modest'' and would amount, in the estimation of IPCC, to not more than 3 percent of the global GDP in 2030. In other words, the level of prosperity that the world would have reached without mitigation would at worst be postponed by a few month or a year at the most. This is clearly not a very high price to pay for protecting hundreds of millions of people from the worst risks associated with climate change. Any such effort , however, would require lifestyles to change appropriately also. Mitigation of green house gas emission is not a mere technological fix. And clearly requires changes in lifestyles and transformation of a country's economic structure, whereby effective reduction in emissions is brought about, such as through the consumption of much lower quantities of animal protein, The Food and Agriculture Organization (FARO) has determined that the emissions from the livestock sector amount to 18 percent of the total. The reduction of emissions from this Source is entirely in the hands of human beings, who have never questioned the impacts that their dietary habits of consuming more and more animal protein are bringing about. Mitigation overall has huge co benefits, such as lower air pollution and health benefits, higher energy security and greater employment

8. According to the passage, which of the following would help in the mitigation of greenhouse gases
1. Reducing the consumption of meat
2. Rapid economic liberalization
3. Reducing the consumption
4. Modern management practices of livestock

Select the correct answer using the code given below
(a) 1, 2 and 3 (b) 2,3 and 4
(c) 1 and 3 only (d) 2 and 4

Ans. (c) According to the passage, mitigation would require a major shift in the way we produce and consume energy. Mitigation of green house gas is not a merely a technological fix. And clearly requires changes in lifestyle. Whereas effective reduction in emissions is brought about, such as through the consumption of much cover quantities of protein. So it is very clear that reducing the consumption of meat and reducing the consumption. Thus, option (c) is correct.

9. Why do we continue to depend on the fossil fuels heavily ?
1. Inadequate technological development
2. Inadequate funds for research and development
3. Inadequate availability of alternative sources of energy

Select the correct answer using the code given below :
(a) only 1 (b) 2 and 3
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) According to the passage, it has been stated that in spite of large climate change patterns happening, a desirable shift away from fossil fuel-based sources of energy is not happening is India.
It can be attributed to low-level of technological advancement to develop non-fossil fuel-based sources of energy, this is because of inadequate funds for research and development provided by the government to develop technology for cleaner sources of energy. These lead to inadequate availability of alternative sources of energy in the country.
Hence, the correct option is 1,2 and 3.

10. According to the passage, how does the mitigation of greenhouse gases help us ?
1. Reduces expenditure on public health
2. Reduces dependence on livestock
3. Reduces energy requirements
4. Reduces rate of global climate change

Select the correct answer using the code given below
(a) 1, 2 and 3
(b) 1, 3 and 4
(c) 2, 3 and 4
(d) 1 and 4 only

Ans. (b) According to the passage, it has been highlighted that mitigation of climate change has the following co-benefits.
• Improved public health, leading to reduced expenditure on health.
• Reduced rate and impact of climate change.
• Energy security, a dimension of which is demand-side management i.e. reducing energy requirements.
Hence, the correct answer is 1, 3 and 4.

11. **What is the essential message of the passage.**
(a) We continue to depend on fossil fuels heavily
(b) Mitigation of the greenhouse gases is imperative
(c) We must invest in research and development/
(d) People must change their lifestyle

Ans. (b) According to the passage, the essential message is mitigation of the greenhouse gas is imperative.

Directions for the following 8 (Eight) items :
Read the following two passages and answer the items that follow each passage. Your answer to these items should be based on the passages only.

Passage-1

The Himalayan ecosystem is highly vulnerable to damage, both due to geological reasons and on account of the stress caused by increased pressure of population, exploitation of natural resources and other related challenges. These aspects may be exacerbated due to the impact of climate change. It is possible that climate change may adversely impact the Himalayan ecosystem through increased temperature altered precipitation patterns, episodes of drought and biotic influences. This would not only impact the very sustenance of the indigenous communities in uplands but also the life of downstream dwellers across the country and beyond, Therefore, there is an urgent need for giving special attention to sustain The Himalayan ecosystem , This would require conscious efforts for conserving all the representative systems.

Further, it needs to be emphasized that the endemics with restricted distribution, and most often with specialized habitat requirements are among the most vulnerable elements. In this respect the Himalayan biodiversity hot spot, with rich endemic diversity, is vulnerable to climate change. The threats include possible loss of genetic resources and species, habitats and concomitantly a decrease in ecosystems services. Therefore, conservation of endemic elements in representative ecosystems/habitats assumes a great significance while drawing conservation plans for the region.

Towards achieving the above, we will have to shift towards contemporary conservation approaches, which include a paradigm of landscape level interconnectivity between protected area systems. The concept advocates a shift from the species-habitat focus to an inclusive focus on expanding the biogeography range so that natural adjustments to climate change can proceed without being restrictive.

12. Consider the following statements:
According to the passage, the Adverse impact of climate change on the ecosystem can be a
1. Permanent disappearance of some of its flora and fauna.
2. Permanent disappearance of ecosystem itself.

Which of the statements given above is/are correct?
(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (a) According to the passage, it is very possible that climate change may adverse impact the Himalayan ecosystem through increased temperature altered precipitation patterns, episode of drought and biotic influence thus it will impact on flora and fauna and sometime permanent disappearance of flora and fauna will take place. Whereas permanent disappearance of ecosystem is not possible thus only statement (1) is correct and statement (2) is wrong. So option (a) is correct.

13. Which one of the following statements best implies the need to shift toward contemporary conservation approach?
(a) Exploitation of natural resources causes a stress on the Himalayan ecosystem.
(b) Climate change alters precipitation patterns. Causes episodes of drought and biotic interference.
(c) The rich biodiversity, including endemic diversity, makes the Himalayan region a biodiversity hotspot.
(d) The Himalayan biogeographic region should be enabled to adapt to climate change smoothly

Ans. (d) According to the passage, the Himalayan biogeographic region should be enabled to adapt to climate change smoothly is in very much need to shift contemporary conservation approach.

14. What is the most important message conveyed by the passage?
(a) Endemism is a characteristic feature of Himalayan region.
(b) Conservation efforts should emphasize on biogeographic ranges rather than on some species or habitats.
(c) Climate change has adverse impact on the Himalayan ecosystem.
(d) Without Himalayan ecosystem, the life of the communities of uplands and downstream will have no sustenance